Class 5

Experimental High-Energy Physics

Detectors, Accelerators, and HEP-Specific Techniques

Duration: 3 months · 13 weeks · ~5–10 hrs/week
Part of the 2026–2027 Self-Study Plan · RHI → HEP Transition

Starts from what you already know as an RHI experimentalist and builds toward HEP-specific territory: hadron-collider triggers, pileup, luminosity, and future-collider detector design.

Course Overview

As a RHI experimentalist, you know detectors. You have worked with tracking, calorimetry, and PID. This class does not re-teach what you already do; it fills the HEP-specific gaps and reframes your existing knowledge for HEP work. The emphasis is on the parts of collider experimentation that differ between the heavy-ion context and the pp/e+e- HEP context: luminosity rather than centrality, pileup rather than event-by-event multiplicity, trigger rates of 40 MHz rather than kHz, and the statistics culture of precision measurement and search.

Read two modern TDRs end-to-end (Weeks 8 and 13). Pick ones you could imagine working on: e.g., ATLAS ITk, CMS HGCAL, sPHENIX Tracking, or a Higgs-factory detector proposal.

The accelerator portion (Weeks 10–12) is compressed but crucial — HEP collaborators will assume you understand what luminosity and beam quality actually mean.

Learning Goals

- Frame your RHI detector knowledge in HEP language, including its HEP-specific limitations.
- Understand silicon-tracker physics and the scaling challenges of HL-LHC.
- Understand calorimetry, including particle flow and the recent shift toward highly granular calorimetry.
- Explain hadron-collider triggering from L1 through HLT to offline.
- Understand LHC accelerator physics at the level of: why is the luminosity what it is? What limits it?
- Read and critically review a detector TDR.

Prerequisites & Background Check

You already have most of the experimental background. Gaps are more likely on the HEP-statistics and accelerator side.

- Working RHI experimental background.
- E&M including radiation and multipoles.
- Familiarity with ROOT or Python + uproot.
- Basic statistics: likelihood, chi-squared, hypothesis testing.

Textbooks & Core Resources

Primary Textbooks

- **Grupen & Shwartz**, *Particle Detectors*, 2nd ed. — main detector-physics text.
- **Wiedemann**, *Particle Accelerator Physics*, 4th ed. — accelerator reference.
- **Thomson**, *Modern Particle Physics* — read for HEP phenomenology context; Chs. 1, 4, 5 especially.

Supplementary / Reference

- **Leo**, *Techniques for Nuclear and Particle Physics Experiments* — review for fundamentals.
- **Knoll**, *Radiation Detection and Measurement* — electronics and signal processing.
- **Blum, Riegler & Rolandi**, *Particle Detection with Drift Chambers*.
- **S.Y. Lee**, *Accelerator Physics* — alternative to Wiedemann; more pedagogical.
- **Cowan**, *Statistical Data Analysis* — the HEP statistician's reference.

Online Lectures & Notes

- PDG reviews: “Passage of Particles Through Matter” and “Particle Detectors at Accelerators.”
- CERN Accelerator School (CAS) proceedings — free on CDS.
- US Particle Accelerator School (USPAS) lecture materials.
- ATLAS, CMS, ALICE, LHCb TDRs and Phase-II upgrade documents.
- sPHENIX TDRs — good bridge given your background.
- Snowmass 2021 (and subsequent) detector and accelerator working group reports.
- ESPP 2026 update documents.

Key Papers & Reviews

- ATLAS and CMS *JINST* detector papers.
- Wigmans, *Calorimetry for Particle Physics* (Rev. Mod. Phys.).
- Particle Flow papers: CMS PF paper (Sirunyan et al., JINST 2017).
- HL-LHC accelerator technical design documents.
- Recent future-collider CDRs: FCC-ee, FCC-hh, ILC, CLIC, C^3, muon collider proposals.

13-Week Schedule

Weeks 1–3 compress what you already know as refresh; real work starts Week 4.

Week	Topic	Readings / Activities	Deliverable
1	Interactions in matter (refresh)	Grupen-Shwartz Ch. 1-2; PDG “Passage of Particles”	Work practice problems
2	Gaseous & silicon trackers	Grupen-Shwartz Ch. 3-4; Blum-Riegler-Rolandi excerpts	Resolution estimates for ATLAS ITk
3	Calorimetry	Grupen-Shwartz Ch. 7-8; Wigmans review	Compare EM vs hadronic resolution
4	Particle ID & particle flow	Grupen-Shwartz Ch. 9; CMS PF paper	Sketch a PF reconstruction chain
5	Trigger & DAQ at hadron colliders	ATLAS/CMS L1/HLT public papers	Trigger menu design exercise
6	Pileup, luminosity, backgrounds	ATLAS/CMS pileup & luminosity notes	Estimate pileup at HL-LHC
7	Statistics & analysis methods in HEP	Cowan Chs. 1-6; PDG statistics review	Likelihood fit exercise in Python
8	TDR deep-dive I	Read one full TDR (e.g., ATLAS ITk or CMS HGCAL)	5-page summary of TDR
9	Accelerator I: beam dynamics	Wiedemann Chs. 1-4	Transverse beam optics problems
10	Accelerator II: RF, longitudinal dynamics	Wiedemann Chs. 5-7	Derive synchrotron eqns
11	Accelerator III: synchrotron radiation, luminosity	Wiedemann Chs. 8-9	Luminosity budget for HL-LHC
12	Future colliders	FCC-ee CDR; muon collider docs; C<super>3</super> docs	Comparison table of proposals
13	TDR deep-dive II; capstone	Read a second TDR from a contrasting experiment	Capstone writeup

Pacing assumes ~5–10 hrs/week on this course while running in parallel with one other class. If time is tighter, cut problem sets first; if time is looser, add problems and deeper paper reading.

Problem Sets & Exercises

Many of the exercises here are mini-design problems rather than textbook problems. That's deliberate — experimental HEP is design.

- Design a silicon pixel layout for a nominal ATLAS ITk layer: given radius, occupancy target, and particle flux, choose pitch and thickness. Justify.
- For a given calorimeter with fixed sampling fraction, estimate energy resolution for 50 GeV electrons and pions.
- Sketch a minimal trigger menu for a pp collider operating at 10^{34} cm^{-2} s^{-1} at L1 and HLT bandwidth budgets; defend your bandwidth allocation.
- Compute expected pileup at $\mu = 200$ for HL-LHC; estimate consequence for jet-mass resolution.
- Likelihood-fit exercise: given a toy background + signal model, extract the signal strength with systematic nuisance parameters.
- Wiedemann problems on closed-orbit correction and betatron tune.
- From the FCC-hh CDR, extract the instantaneous luminosity target and reproduce the estimate of integrated luminosity over 25 years.

Capstone Project

Two-part. Part A is a TDR review; Part B is a personal positioning document.

- **Part A.** Write a 6–8 page critical review of one detector TDR (TDR of your choice among ATLAS ITk, CMS HGCAL, sPHENIX, ALICE 3, or a future-collider proposal). Summarize design, identify the 3–4 most physically interesting tradeoffs, and assess one area where you could personally contribute.
- **Part B.** 2–3 page memo: *What my RHI experimental background buys me for HEP*. Be specific about hardware (TPC, calorimeter operations), software (simulation, reconstruction, correlation-function analysis), and statistical cultures (high multiplicity, event-by-event) — and where the cultures differ.

Milestones & Self-Assessment

At each checkpoint, you should be able to do the listed tasks without looking them up. If you can't, go back rather than pushing forward.

- **End of Week 4:** You can articulate exactly what is the same and what is different between a typical heavy-ion detector and a typical pp collider detector, with numbers.
- **End of Week 7:** You can set up a likelihood fit in Python with nuisance parameters, and explain to a heavy-ion colleague what a “profile likelihood” is.
- **End of Week 11:** You can explain what luminosity is in terms of beam parameters, and identify the 2–3 quantities that limit it at the LHC.

- **End of Week 13:** You have read two full TDRs. You can walk into an ATLAS or CMS detector meeting and follow the discussion.

End of class plan. Iterate after the first two weeks — what was over-ambitious gets cut; what felt thin gets deepened.